

1. State the definition of a group.

A group is a set G on which is defined a binary operation satisfying the following axioms.

- ① Associativity: $a(bc) = (ab)c$ for all $a, b, c \in G$,
- ② There is an identity $e \in G$ for which $ea = a = ae$ for any $a \in G$.
- ③ Each $a \in G$ has an inverse $a^{-1} \in G$ for which $aa^{-1} = e = a^{-1}a$.

2. Write the addition table for the group $\mathbb{Z}_3$.

+	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

3. Give an example of a group with infinitely many elements, having the property that every nontrivial subgroup also has infinitely many elements.

$$(\mathbb{Z}, +) \text{ or } (\mathbb{R}, +), \text{ etc.}$$

4. State three different abelian groups with 8 elements. (You do not have to write out their tables.)

$$\mathbb{Z}_8 \quad \mathbb{Z}_4 \times \mathbb{Z}_2 \quad \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$$

5. Optional extra credit: State two different *nonabelian* groups with 8 elements.

$$D_4 \text{ (symmetries of a square)}$$

$$Q = \{\pm 1, \pm i, \pm j, \pm k\} \text{ (quaternion group)}$$